

Our Data Science Learning Story

These five Learning Intentions guide the course. The Success Criteria follow the course's instructional topics so each routine assignment has one clear evidence target. Related note and lab sets are combined only when they develop the same enduring capability.

1 LI 1 COMPUTATIONAL PROBLEM SOLVING

Represent, control, test, and solve

2 LI 2 BUILDING TRUSTWORTHY DATA

Acquire, clean, transform, and combine

3 LI 3 DISCOVERING & EXPLAINING PATTERNS

Summarize, visualize, compare, and communicate

DATA SCIENCE WITH PYTHON
DATA SCIENTIST
code | prepare | explore | infer | model

4 LI 4 REASONING UNDER UNCERTAINTY

Model chance, sample, test, and conclude

5 LI 5 MODELING & PREDICTION

Build, evaluate, predict, and explain limits

I am becoming a data scientist who can transform raw data into defensible insights and informed predictions.

LI 1 COMPUTATIONAL PROBLEM SOLVING

I am learning to use Python to represent information, control program flow, and build reliable solutions.

LI 2 BUILDING TRUSTWORTHY DATA

I am learning to acquire, organize, clean, transform, and combine data for trustworthy analysis.

LI 3 DISCOVERING & EXPLAINING PATTERNS

I am learning to summarize and visualize data to identify patterns, examine relationships, and explain what they reveal.

LI 4 REASONING UNDER UNCERTAINTY

I am learning to use probability, sampling, and inference to evaluate evidence and draw defensible conclusions.

LI 5 MODELING & PREDICTION

I am learning to build, interpret, and evaluate models to make informed predictions and explain their limitations.

HOW TO READ THE TILES Each numbered tile is one topic-based Success Criterion. Its badges show the Learning Intentions it supports, and its Primary Evidence line identifies the notes and lab that normally connect to it. Projects are intentionally handled separately because they integrate learning across several criteria.

Success Criteria

Each criterion is a topic-based evidence target. Routine assignments normally connect to one primary criterion; the Primary Evidence line shows where that learning is developed.

LI 1 Computational Problem Solving

LI 2 Building Trustworthy Data

**LI 3 Discovering & Explaining
Patterns**

LI 4 Reasoning Under Uncertainty

LI 5 Modeling & Prediction

SC 0

I can use Jupyter notebooks, GitHub, and version control to organize, track, and submit my data-science work.

PRIMARY EVIDENCE Lab: GitHub Workshop

LI 1

SC 1

I can use Python functions and print() parameters to produce clear, correctly formatted output.

PRIMARY EVIDENCE Notes/Lab: The Print Function

LI 1

SC 2

I can interpret Python literals and identify the data type of a value.

PRIMARY EVIDENCE Notes/Lab: Literals

LI 1

SC 3

I can use variables, type conversion, and operators to represent, update, and compute with data.

PRIMARY EVIDENCE Notes: Operators, Variables | Lab: M & M Statistics

LI 1

SC 4

I can create and evaluate Boolean expressions to compare values and filter data.

PRIMARY EVIDENCE Notes: Booleans | Lab: Netflix Statistics

LI 1

SC 5

I can use if, elif, else, and nested conditionals to control decisions in a program.

PRIMARY EVIDENCE Notes/Labs: Conditionals Parts 1-2

LI 1

SC 6

I can create, access, modify, sort, and process data stored in one-dimensional lists.

PRIMARY EVIDENCE Notes/Labs: Introduction to Lists; Working with Lists

LI 1 LI 2

SC 7

I can organize, access, modify, and combine data stored in rows and columns using two-dimensional lists.

PRIMARY EVIDENCE Notes/Lab: Two-Dimensional Lists

LI 1 LI 2

SC 8

I can use for, while, nested loops, and loop controls to repeat processes and traverse data.

PRIMARY EVIDENCE Notes/Labs: Loops Parts 1-2

LI 1

SC 9

I can define, call, and test reusable functions using parameters, return values, and appropriate variable scope.

PRIMARY EVIDENCE Notes/Labs: Functions Parts 1-2

LI 1

SC 10

I can acquire, read, write, and convert data from APIs and common file formats, including text, CSV, and JSON.

PRIMARY EVIDENCE Notes: Data Acquisition; Python Files | Labs: Data Acquisition; Working With Files

LI 2

SC 11

I can parse, clean, transform, and format text data using string operations and methods.

PRIMARY EVIDENCE Notes: Python Strings; String Methods | Lab: String Operations

LI 1 LI 2

SC 12

I can create, access, update, and process key-value data using dictionaries.

PRIMARY EVIDENCE Notes: Creating/Using Dictionaries | Lab: Dictionaries

LI 1 LI 2

Success Criteria, continued

The later criteria follow the course's progression from preparing data to explaining patterns, reasoning under uncertainty, and building models.

LI 1 Computational Problem Solving

LI 2 Building Trustworthy Data

LI 3 Discovering & Explaining
Patterns

LI 4 Reasoning Under Uncertainty

LI 5 Modeling & Prediction

SC 13

I can create, load, inspect, select, filter, and modify pandas DataFrames.

PRIMARY EVIDENCE Notes: DataFrames I-II | Lab: Manipulating Data with Pandas

LI 2

SC 14

I can group data, compute aggregate statistics, and create pivot tables to compare results.

PRIMARY EVIDENCE Notes/Lab: Aggregate Statistics with Pandas

LI 2 LI 3

SC 15

I can merge, concatenate, and analyze related data stored across multiple tables.

PRIMARY EVIDENCE Notes/Lab: Multiple Tables with Pandas

LI 2 LI 3

SC 16

I can classify, convert, order, and encode numerical and categorical variables for analysis.

PRIMARY EVIDENCE Notes: Variable Types for Data Science | Lab: Managing Data Types

LI 2

SC 17

I can summarize and visualize a single feature using appropriate measures of center, spread, and distribution.

PRIMARY EVIDENCE Notes/Lab: Summarizing Single Features

LI 3

SC 18

I can summarize, visualize, and interpret relationships between features using methods appropriate to their variable types.

PRIMARY EVIDENCE Notes: Relationships by Variable Type | Lab: Summarizing Relationships Between Features

LI 3

SC 19

I can use probability rules, simulation, and probability distributions to model random events and calculate probabilities and expected outcomes.

PRIMARY EVIDENCE Notes: Probability Rules; Distributions; Poisson | Labs: Probability Distributions; Poisson

LI 4

SC 20

I can build and interpret sampling distributions, apply the Central Limit Theorem, and evaluate bias and standard error.

PRIMARY EVIDENCE Notes/Lab: Sampling Distributions

LI 4

SC 21

I can test a claim about a population mean by defining hypotheses, calculating a p-value, and drawing a defensible conclusion.

PRIMARY EVIDENCE Notes: Mean Hypothesis Testing | Lab: Climate Claims with Hypothesis Testing

LI 4

SC 22

I can use simulation and bootstrap methods to test claims about proportions and medians and interpret one- and two-sided p-values.

PRIMARY EVIDENCE Notes: Binomial/Median Hypothesis Testing | Lab: Best Metro to Launch Adult Life

LI 4

SC 23

I can fit, interpret, evaluate, and use a linear regression model to explain relationships and make predictions.

PRIMARY EVIDENCE Notes: Linear Regression | Lab: Work, Wealth, and the Future of the Workweek

LI 3 LI 5

SC 24

I can identify features and labels, distinguish learning types, and choose an appropriate machine-learning task for a stated purpose.

PRIMARY EVIDENCE Notes: Introduction to Machine Learning | Lab: Penguin Patterns

LI 5

SC 25

I can prepare data, build and evaluate an introductory machine-learning model, and explain its predictions and limitations.

PRIMARY EVIDENCE Planned: Machine-learning model-building notes and lab (2-3 weeks)

LI 2 LI 5